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Machine learning based real-time assessment of fabrication deviation induced mechanical performance variations in stretchable silicon arrays

Bo Wen¹, Han Xu¹✉, Yikang Ding², Qi Wang², Pan Zhang¹ , Chi Zhang¹ and Wei Wang^{1,3,4} ✉

Abstract

Microelectromechanical system fabrication represents a promising approach for silicon-based flexible electronics, leveraging its scalability and miniaturization merits. However, fabrication-induced geometric deviations stretchable microstructures can result in significant variations in mechanical performances. Current assessment methods lack sufficient accuracy for these precision-sensitive manufacturing processes. This work proposes a machine-learning (ML)-based assessment methodology for accurately and rapidly predicting the mechanical performances, including equivalent Young's modulus and the maximum elastic stretchability, of Parylene three-dimensional micro-Kirigami stretchable structures in a stretchable silicon array affected by the fabrication-induced geometric deviations. By applying the dimensionality reduction technique specifically designed for few-shot ML modeling, the framework achieves prediction accuracies exceeding 95% on the test set. SHapley Additive exPlanations (SHAP) analysis is further utilized to quantify the impact of various geometric features on mechanical performances. This ML-based assessment methodology successfully facilitates real-time feedback from process-induced geometric deviations to the qualification probability of mechanical performances. This proposed approach supports design-for-manufacturability (DFM) of silicon-based stretchable arrayed devices manufacturing and lays the foundation for high-consistency wafer-scale manufacturing of high-performance stretchable silicon electronics.

Introduction

Flexible electronics, characterized by their deformable, stretchable and bendable performance, demonstrate significant advantages over conventional rigid electronic components in adapting to dynamic conformal requirements on arbitrarily curved surfaces. This unique adaptability enables broader application potentials across diverse fields including wearable devices¹, medical monitoring systems², soft robotics³, and AR/VR technologies^{4,5}.

Current flexible and stretchable electronics manufacturing strategies primarily fall into two categories. The first approach employs flexible materials to fabricate

functional devices for computing, memory, and sensing applications^{6,7}. While devices fabricated entirely from flexible materials exhibit intrinsically favorable mechanical performance, their electrical performance can hardly match their silicon-based counterparts⁸. The second strategy leverages the electrical performance of silicon electronics and mechanically stretchable interconnects to achieve high-performance silicon-based flexible electronics⁹. These devices combine the electrical performance advantages of silicon electronics, typically realized by wafer-scale thinning of silicon devices and transferring them onto flexible substrates containing stretchable electrical interconnects^{10–12}. However, the mechanical and electrical interconnects on flexible substrates are difficult to be high-consistent miniaturized due to the fabrication methods¹³, impeding massive production in industry. Furthermore, the relatively large feature sizes of the stretchable structures in the transfer-based silicon electronics create interdependent trade-offs between mechanical

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compliance and electrical performance¹⁴. Consequently, wafer-scale fabrication approaches for mechanical and electrical interconnects emerges as a promising pathway to achieve high-performance silicon-based flexible electronics industrialized manufacturing^{15–17}.

To achieve enhanced stretchability, the feature size of mechanical interconnecting structures can be reduced to the 10 μm . However, as feature sizes decrease, the impact of inevitable fabrication-induced geometric deviations on mechanical performances becomes increasingly pronounced. There is still a lack of efficient assessment methods for the impact of geometric deviations introduced during the microelectromechanical system (MEMS) fabrication of stretchable microstructures on the mechanical performance of stretchable silicon array devices.

The relationship between stretchable microstructure feature size and mechanical performance has been extensively studied in stretchable silicon array. Both analytical models and finite element analysis (FEA) simulations are widely used to correlate the feature size with the mechanical performance in representative stretchable structures (e.g., serpentine, fractal, and Kirigami)^{18–21}. While analytical models enable rapid computation with negligible computational costs, their accuracy is limited by inherent assumptions and neglect of non-ideal effects. FEA simulations can achieve higher accuracy but suffer from computational inefficiency, particularly for large scale complex stretchable microstructures. In addition, experimental tensile testing remains the primary method for characterizing mechanical performances of stretchable arrays^{22–24}. However, such mechanical testing cannot achieve real-time or non-destructive data acquisition.

machine-learning (ML)-based methods offer significant potential for overcoming limitations in accuracy, efficiency and non-destruction, to predict mechanical performance. Pankaj Rajak et al.²⁵ use reinforcement learning to generate a wide range of highly stretchable MoS_2 kirigami structures. After training by molecular dynamics simulation data, the RL agent proposes 6-cut kirigami structures that have stretchability above 45%. Rui et al.^{26,27} employed ML to investigate the impacts of feature sizes on maximum strain in serpentine structures and identified the most influential feature size. However, these studies focus on the structural optimization of stretchable microstructures, and the consideration of fabrication-induced geometric deviations of stretchable microstructures is insufficient.

In this work, we propose a ML-based assessment method to evaluate the effects of fabrication-induced geometric deviations on the mechanical performance of stretchable silicon arrays fabricated by MEMS processing techniques. We introduce a ML framework tailored for

predicting the mechanical performance of stretchable silicon arrays, specifically addressing variations in the equivalent Young's modulus and the maximum elastic stretchability due to fabrication-induced deviations. We demonstrate the feasibility of our method by using a Parylene three-dimensional Kirigami-like stretchable silicon array (Parylene 3D KiSS array) as an illustrative example. A customized ML model is developed to establish an efficient mapping between feature size and mechanical performances (the equivalent Young's modulus and the maximum elastic stretchability) in stretchable silicon arrays. A dimensionality reduction technique is implemented to enhance accuracy and mitigate overfitting risks in few-shot ML modeling. The most influential feature size on the mechanical performance is identified through ML model-based SHapley Additive exPlanations (SHAP) analysis. This methodology is dedicated to rapid evaluate fabrication induced mechanical performance variations in stretchable microstructures interconnected stretchable silicon arrays, further realize the real-time prediction of the mechanical performances during and post-fabrication, supporting high-consistency manufacturing of stretchable silicon arrays.

Results and discussions

The workflow of the ML-based assessment of fabrication deviation induced mechanical performance variations in stretchable silicon array is schematically presented in Fig. 1. The workflow includes three stages: data preparation, ML modeling, and model-assisted fabrication development. This work starts with the preparation of the Parylene 3D KiSS array. During the data preparation stage, optical microscope (OM)-based CD measurement is performed on the as-fabricated Parylene 3D Kirigami-like stretchable microstructures for rapid measurement of the feature sizes, including adjacent Parylene beam spacing w , slit length l , minimum line width λ , and adjacent slit spacing s . Uniaxial tensile tests are conducted on the as-fabricated Parylene 3D KiSS array to obtain stress-strain curves. Two mechanical performance of behaviors, the equivalent Young's modulus k_{eff} and the maximum elastic stretchability $\varepsilon_s^{\text{max}}$, are extracted from the stress-strain curve. During the ML modeling stage, two distinct frameworks are systematically implemented: a multilayer-perceptron (MLP) framework and a tree-based CatBoost framework. A performance-driven selection protocol is subsequently executed to identify the optimally configured model with the highest prediction accuracy. During model-assisted fabrication development stage, we demonstrate two key functions of our ML-based assessment methodology in facilitating batch manufacturing of Parylene 3D KiSS arrays: (1) critical feature size extraction and (2) wafer-level assessment of qualification probability during fabrication. The subsequent sections will provide

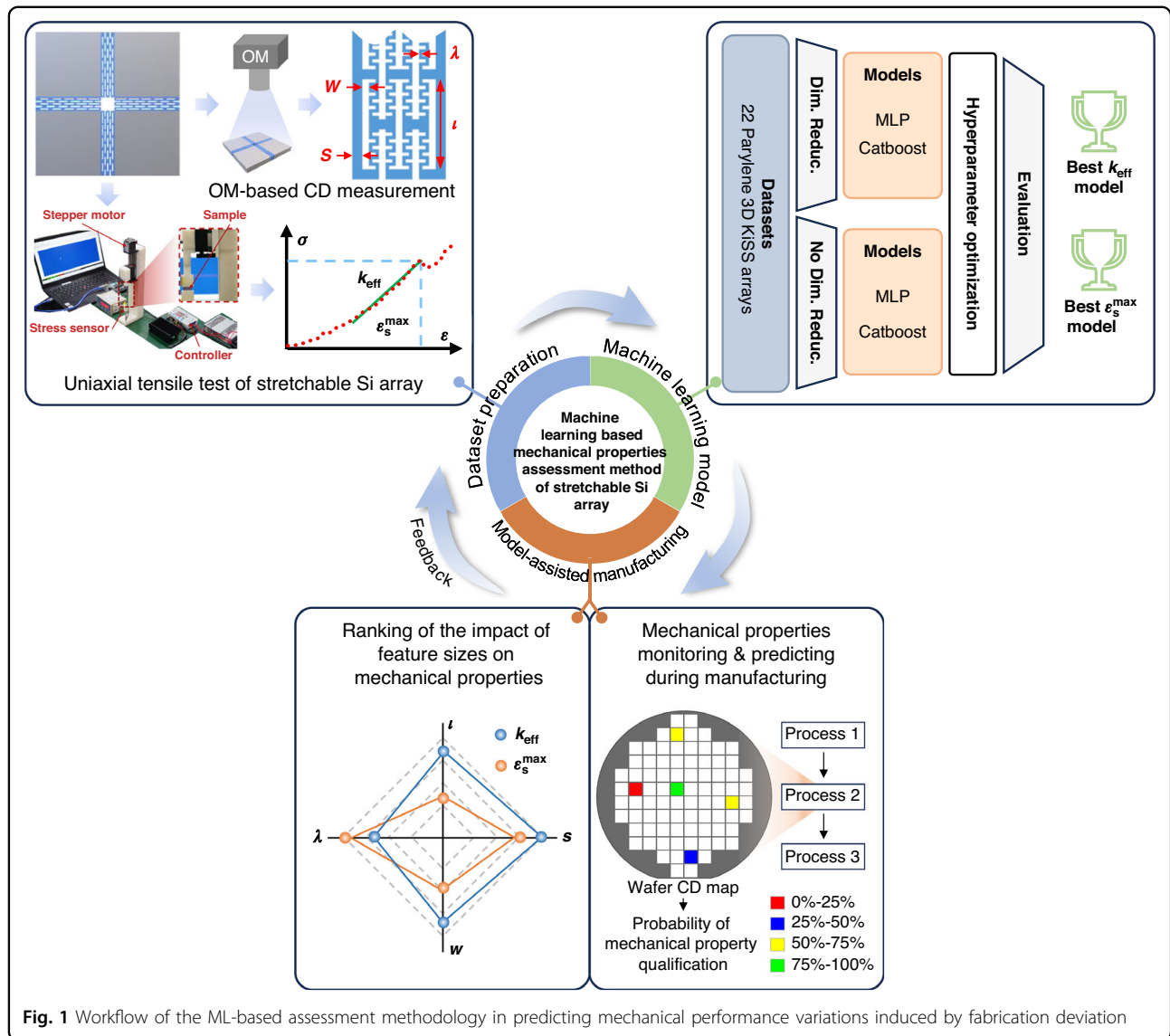


Fig. 1 Workflow of the ML-based assessment methodology in predicting mechanical performance variations induced by fabrication deviation

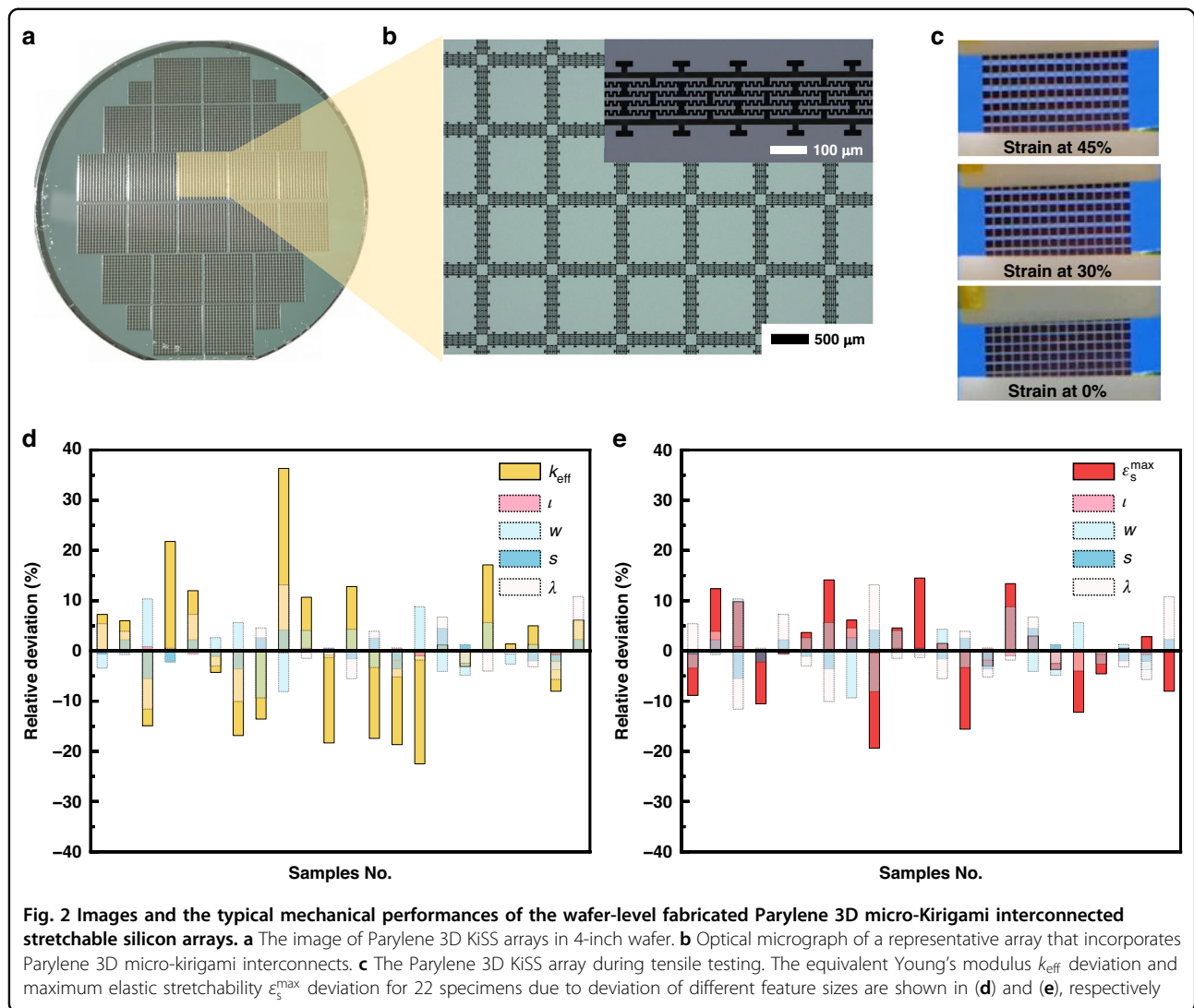
comprehensive delineations of each stage within the assessment methodology.

Data preparation

The Parylene 3D KiSS arrays are prepared through a wafer-level microfabrication process with MEMS and post-CMOS fabrication techniques. Figure 2a, b shows the stretchable silicon arrays in a 4-inch wafer in-process. The feature sizes of the Parylene 3D Kirigami-like microstructures are defined by photolithography and subsequent deep reactive ion etching (DRIE). The feature sizes adopted for mechanical performance assessment are measured post-etching by optical microscope. Figure 2c shows the stretchable silicon array under tensile testing. The equivalent Young's modulus k_{eff} and the maximum elastic stretchability ϵ_s^{max} are calculated from the stress-

strain curves obtained via tensile tests. To build the ML dataset, we randomly select specimens from three wafers produced in the same batch, yielding 22 valid samples that are subsequently divided into training and testing sets.

Figure 2d, e showed the deviations of the equivalent Young's modulus and the maximum elastic stretchability, respectively, of these stretchable silicon arrays. The results were compared to the average values. The data reveal substantial dispersion: the equivalent Young's modulus exhibited a maximum deviation of 36.3% from its mean and the maximum elastic stretchability displayed a maximum deviation of 19.4% from its mean. Notably, the maximum deviation in any geometric feature sizes in the corresponding Parylene 3D Kirigami-like microstructures reaches only 9.36%, indicating a pronounced amplification effect of feature size deviations on mechanical

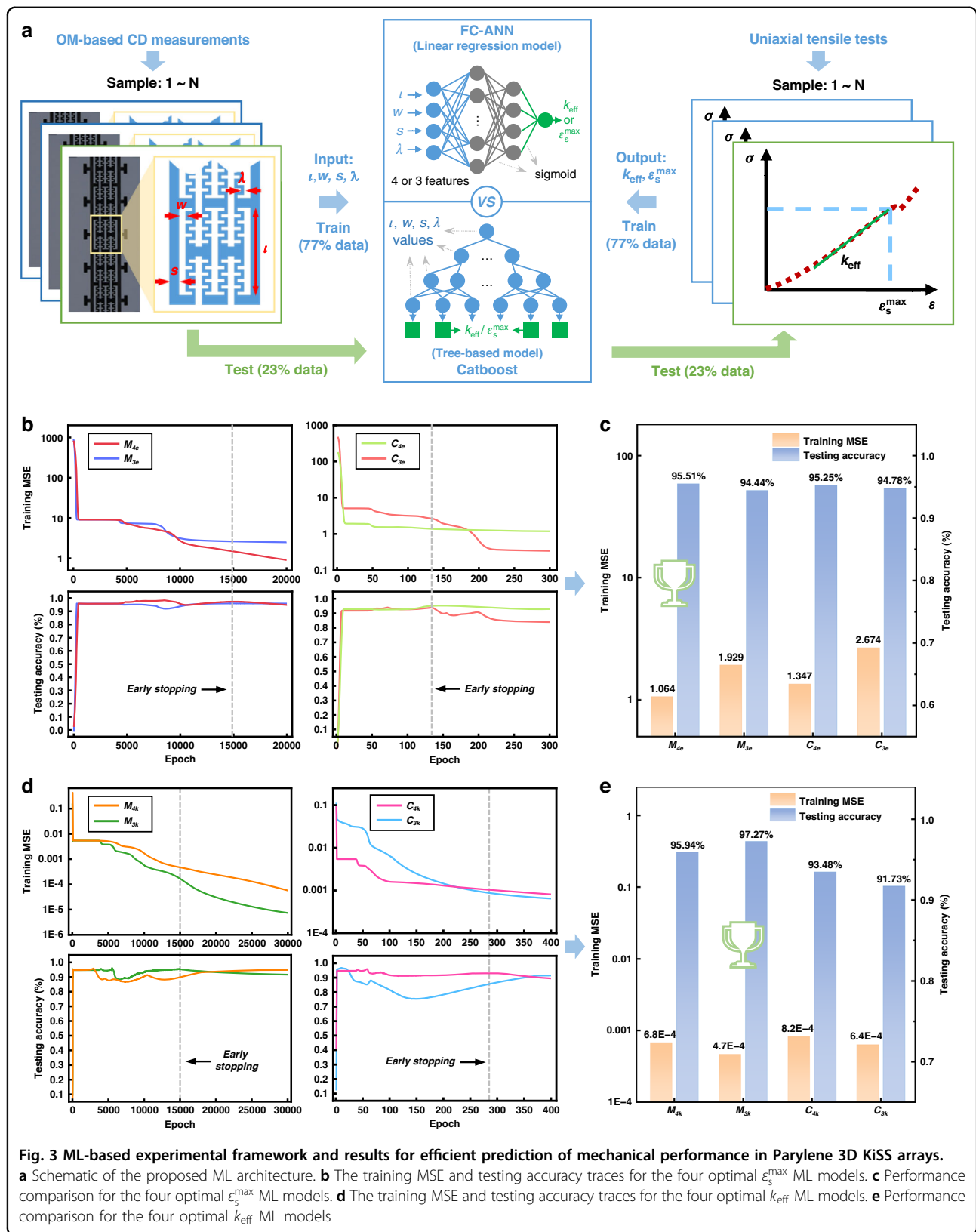


performance variations. This three- to four-fold mismatch underscores a strong amplification effect, whereby small fabrication-induced geometric deviations translate into large shifts in mechanical performances. Therefore, establishing an accurate prediction model correlating feature sizes to mechanical performances is essential for the assessment of fabrication deviations in Parylene 3D KiSS array.

ML modeling

Establishing ML models that generalize well on limited samples is challenging, especially considering the non-linear relationships between feature size and mechanical performances k_{eff} & ϵ_s^{max} . Consequently, we employ two frameworks: a MLP and a tree-based CatBoost model, both have been proven effective for small-sample regression analyses^{28–30}. The architecture of the proposed model is illustrated in Fig. 3a.

Let $\mathbf{x}_k \in \mathbb{R}^k$ denotes the geometric parameter vector characterizing the Parylene 3D Kirigami-like microstructures, where $\mathbf{x}_k = [x_1, x_2, \dots, x_k]^T$ with each element $x_i (i = 1, 2, \dots, k)$ representing a specific feature size. This parameter vector serves as the input parameters of the ML models. The equivalent Young's modulus k_{eff} and the maximum elastic stretchability ϵ_s^{max} are defined as the output parameters of the ML models. The Parylene 3D KiSS microstructure exhibits mechanical behavior that is primarily governed by relative geometrical proportions rather than absolute dimensions. In particular, both the slit length l and the minimum line width λ scale naturally with the adjacent slit spacing s , meaning that the ratios l/s and λ/s capture the key deformation-relevant geometry. The in-plane stretching behavior of this microstructure originates from beam bending, which implies that the ratio of l/s is capable of describing the size-dependent characteristics of the microstructure. The ratio λ/s is



utilized for dimensionality reduction as it is defined by the proportion of the real feature size of the beam to the slits spacing s . In contrast, the adjacent Parylene beam spacing w does not form a meaningful proportional relationship with s , and therefore retaining w in its absolute form preserves essential geometric information. Based on these considerations, dimensionality reduction is implemented, transforming the original four-dimensional feature space ($\mathbf{x}_4 = [l, s, w, \lambda]^T$) into a three-dimensional feature space ($\mathbf{x}_3 = [l/s, w, \lambda/s]^T$). We think such a dimensionality reduction can not only decrease effective degrees of freedom but also suppress spurious correlations among geometric parameters to mitigate overfitting in the few-shot ML modeling setting. On this basis, considering the substantial scale difference between k_{eff} and $\varepsilon_s^{\text{max}}$, we construct four MLP models and four CatBoost models to represent the unknown relationship between mechanical performances k_{eff} & $\varepsilon_s^{\text{max}}$ and the statistical variables $\mathbf{x}_3$ & $\mathbf{x}_4$. The input-output configurations of the eight ML models are schematically summarized in Table 1. The M_{4e} model represents a four-dimensional input MLP framework with the maximum elastic stretchability $\varepsilon_s^{\text{max}}$ as its output. Analogously, the C_{3k} model constitutes a three-dimensional input Catboost configuration designed to predict the equivalent Young's modulus k_{eff} . The accuracy of the ML models is quantified by evaluating mean squared errors (MSE) and mean accuracy. The calculation formulas are as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (1)$$

$$\text{mean accuracy} = 1 - MAPE = 1 - \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (2)$$

Table 1 Input-output configurations of the eight ML models

Model name	Framework	Inputs	Outputs
M_{4e}	MLP	l, w, s, λ	$\varepsilon_s^{\text{max}}$
M_{3e}	MLP	$l/s, w, \lambda/s$	$\varepsilon_s^{\text{max}}$
C_{4e}	Catboost	l, w, s, λ	$\varepsilon_s^{\text{max}}$
C_{3e}	Catboost	$l/s, w, \lambda/s$	$\varepsilon_s^{\text{max}}$
M_{4k}	MLP	l, w, s, λ	k_{eff}
M_{3k}	MLP	$l/s, w, \lambda/s$	k_{eff}
C_{4k}	Catboost	l, w, s, λ	k_{eff}
C_{3k}	Catboost	$l/s, w, \lambda/s$	k_{eff}

where n is the total number of samples in the dataset, $\hat{y}_i$ is the predicted value of the i -th sample, y_i is the observed value of the i -th sample.

All the eight models are optimized using a systematic grid-search procedure to identify the best-performing hyperparameters. For tree-based CatBoost models, the tuned hyperparameters include the number of estimators, maximum tree depth, learning rate, and L2 regularization coefficient for leaf values. For neural-network MLP models, the search space covers the number of hidden layers, neurons per layer, learning rate, and early stopping criterion. For each model, a predefined search space is constructed based on commonly used hyperparameters. Each hyperparameter combination is evaluated by computing the mean accuracy on the testing folds, and the configuration yielding the highest accuracy is selected as the optimal setting. Figure 3b shows the training MSE and testing accuracy curves for the four $\varepsilon_s^{\text{max}}$ prediction models (M_{4e} , M_{3e} , C_{4e} and C_{3e}) under optimal hyperparameter configurations. All training loss curves exhibit gradual reduction and convergence, while testing accuracy curves demonstrate progressive improvement and stabilization, confirming stable training dynamics. After sufficient iterations, all models achieve accuracies exceeding 90%, validating the feasibility of ML for manufacturing parameter prediction. As shown in Fig. 3c, M_{4e} , M_{3e} , C_{4e} , and C_{3e} achieve prediction accuracy values of 95.51%, 94.44%, 95.25%, and 94.78%, respectively. When comparing the MLP model with the CatBoost model, negligible differences are observed in training MSE and testing accuracy, indicating that both frameworks exhibit comparable effectiveness in the $\varepsilon_s^{\text{max}}$ prediction. The four-dimensional models M_{4e} & C_{4e} demonstrate a slight edge over the three-dimensional models M_{3e} & C_{3e} in both training MSE and testing accuracy, suggesting that dimensionality reduction in $\varepsilon_s^{\text{max}}$ ML models might lead to the potential loss of features. Among all four $\varepsilon_s^{\text{max}}$ prediction models, the four-dimensional MLP model M_{4e} exhibits the best performance with the lowest training MSE (1.064) and the highest testing accuracy (95.51%).

Similarly, Fig. 3d shows the training MSE and testing accuracy curves for the four k_{eff} prediction models (M_{4k} , M_{3k} , C_{4k} and C_{3k}) under optimal hyperparameter configurations. As shown in Fig. 3e, M_{4k} , M_{3k} , C_{4k} , and C_{3k} achieved prediction accuracy values of 95.94%, 97.27%, 93.48%, and 91.73%, respectively. When comparing the MLP model with the CatBoost model, the MLP framework demonstrates statistically significant advantages in both training MSE and testing accuracy, suggesting the limitations of the CatBoost framework for k_{eff} prediction. Furthermore, the three-dimensional MLP model M_{3k} exhibits marked superiority over its four-dimensional counterpart M_{4k} across training MSE and testing accuracy, demonstrating the effectiveness of dimensionality

reduction within the MLP framework. Among all four k_{eff} prediction models, the three-dimensional MLP model M_{3k} exhibits the best performance owing to the lowest training MSE (4.7×10^{-4}) and the highest testing accuracy (97.27%).

Therefore, through the systematic implementation of dimensionality reduction and comparative evaluation of two distinct ML frameworks, the optimally configured ML model has been conclusively identified. Models M_{4e} and M_{3k} , exhibiting superior prediction performances, are selected as reliable mapping frameworks for subsequent analysis.

Model-assisted manufacturing development

To quantify how the four geometric features affect the equivalent Young's modulus k_{eff} and the maximum elastic stretchability $\varepsilon_s^{\text{max}}$, SHAP algorithm is applied to extract feature importance from the reliable models M_{4e} and M_{3k} . SHAP algorithm assigns a corresponding SHAP value to each inputting parameter, facilitating a quantitative appraisal of its contribution to the model output³¹. By visualizing the mean absolute SHAP values, the relative importance of each inputting parameter within the model can be ranked. Higher SHAP values indicate stronger impacts on model's prediction values. As illustrated in Fig. 4a, b, results of mean absolute SHAP values reveal that parameter λ dominates $\varepsilon_s^{\text{max}}$, while s exerts the greatest influence on k_{eff} . These findings align with results derived from structural mechanics analytical models¹⁶, validating the accuracy of our ML-based mapping framework. The model-based SHAP analysis fully confirms the powerful capability of our methodology in identifying critical process parameters in stretchable silicon array fabrication. Overall, the SHAP analysis demonstrates that the proposed framework reliably pinpoints the process parameters that dictate mechanical performances in Parylene 3D KiSS array fabrication.

Furthermore, leveraging the Monte Carlo (MC) method for uncertainty quantification^{32–35}, we demonstrate our ML model's utility for quantitative process window analysis in stretchable silicon array fabrication by inputting statistical data with progressively larger standard deviations (σ) into the reliable models M_{4e} and M_{3k} . The detailed workflow is schematically illustrated from Fig. 4c–f. For each feature size, the standard deviations (σ) of all 22 samples are calculated to get the baseline variability. Firstly, one of the feature sizes is selected as the primary parameter. To simulate fabrication-induced geometrical deviations, 100 Gaussian-distributed statistical data sets (σ ranging from 0 to the half of the sample standard deviations) are generated for the primary parameter. Each dataset contains 1000 samples and 100 statistical realizations, forming a three-dimensional array of size $1000 \times 1 \times 100$, as shown in Fig. 4c. An increased σ

corresponds to greater dispersion in the feature size's statistical distribution, which means that the inconsistency in manufacturing becomes stronger.

The remaining three feature sizes are simultaneously sampled from uniform distributions bounded by their measured ranges, forming a $1000 \times 3 \times 100$ array, as shown in Fig. 4d. The combined data sets are passed through our reliable ML model, producing batch predictions of $\varepsilon_s^{\text{max}}$ and k_{eff} . Samples satisfying both $k_{\text{eff}} \in [0.4, 0.5]$ and $\varepsilon_s^{\text{max}} \in [28, 31]$ are classified as qualified; others are deemed unqualified. The yield (Y) is therefore defined as:

$$Y = \frac{N_{\text{qualified}}}{N_{\text{total}}} \quad (3)$$

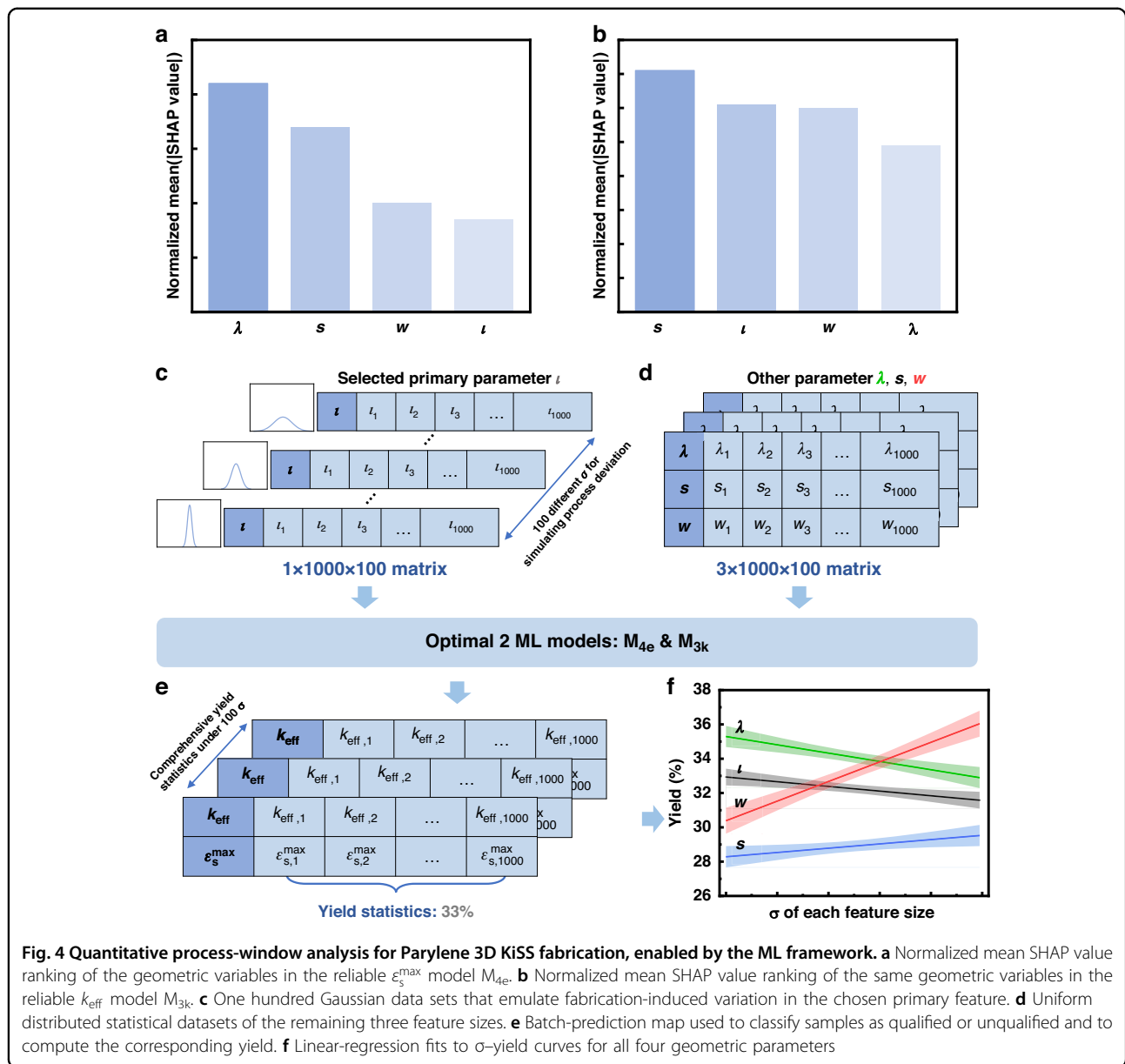
where $N_{\text{qualified}}$ and N_{total} represent the numbers of qualified and total samples of Parylene 3D KiSS arrays, respectively.

The batch prediction outcomes are categorized according to the above criteria, as shown in Fig. 4e. By varying σ of the primary parameter, the relationship between σ and predicted yield (Y) is analyzed quantitatively. Scatter plots are constructed with σ on the x -axis and Y on the y -axis to visualize how increasing geometrical uncertainty influences the expected manufacturing yield. Repeating this process by designating each of the four geometrical parameters as the primary parameter in turn allows evaluation of the σ – Y relationships for all parameters, with linear fits shown in Fig. 4f.

The results reveal that the expected yield exhibits a decline with increasing σ of feature sizes λ and l , aligning with conventional observations of negative correlation between fabrication deviations and yields. However, anomalous positive correlations emerged for feature sizes w and s . We hypothesize that this anomaly is attributed to (1) the midpoint mismatch between the distribution interval and (2) the dual-constraint interplay induced by the simultaneous consideration of $\varepsilon_s^{\text{max}}$ and k_{eff} . Nevertheless, this methodology enables real-time process-window extraction and quantitatively highlights gaps between current process capability and target yield requirements—thereby guiding tolerance allocation in Parylene 3D KiSS array manufacturing.

Model-assisted in-line assessment of mechanical performances

Based on the aforementioned models for real-time prediction of the equivalent Young's modulus k_{eff} and the maximum elastic stretchability $\varepsilon_s^{\text{max}}$, we demonstrate how the framework estimates qualification probability during Parylene 3D KiSS fabrication. As depicted in Fig. 5, in-line metrology acquires three geometric parameters— l , w , and s —while λ remains unmeasured, reflecting scenarios in which

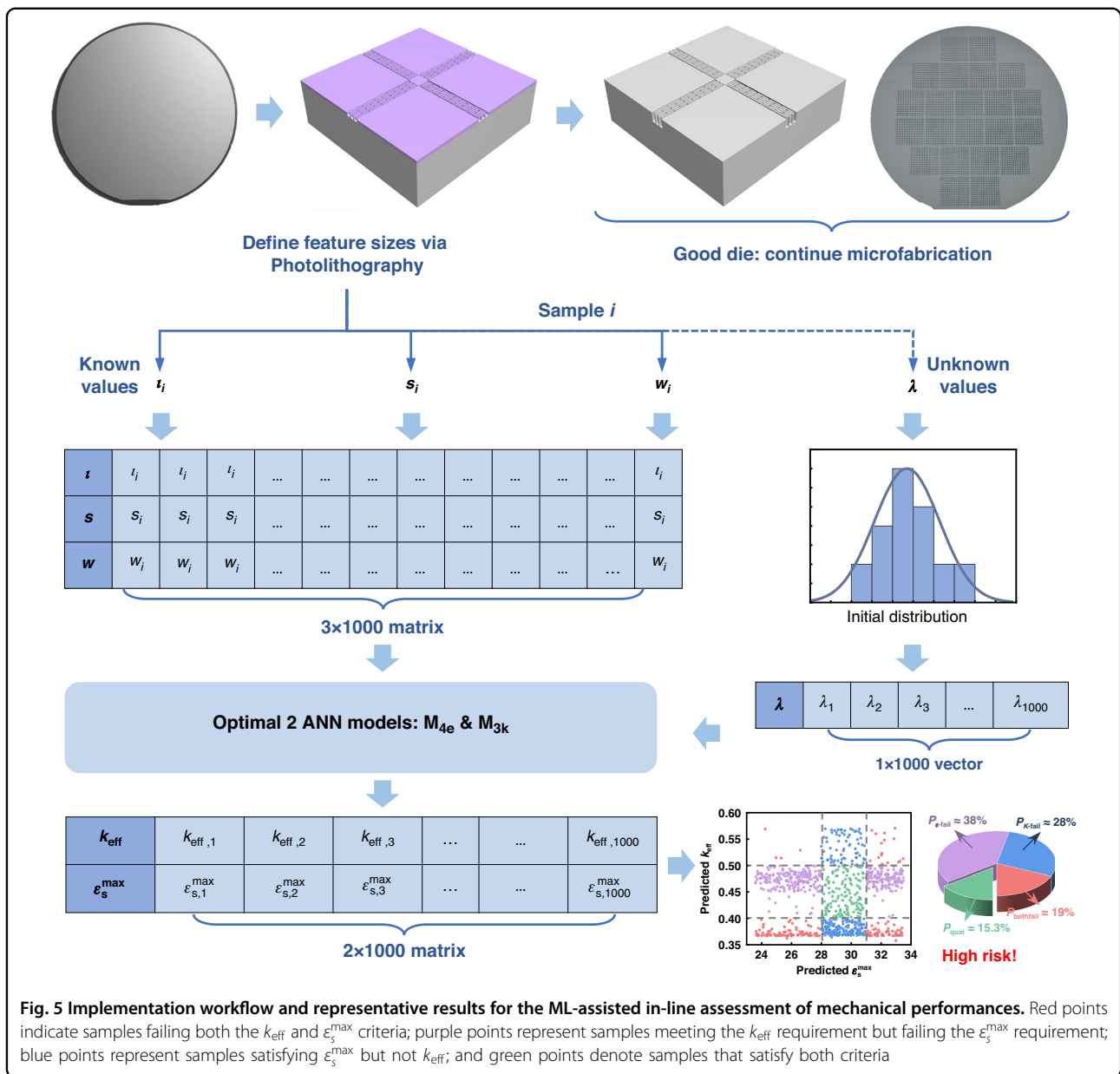


not every feature can be captured within a single process step. By treating λ as a random variable and propagating its anticipated distribution, together with the measured values of l , w , and s , through the trained models, we compute the probability that each in-process array will satisfy the dual specifications on k_{eff} and $\varepsilon_s^{\max}$. This capability enables real-time, sample-specific risk screening and adaptive process control in the production line.

Following the same strategy used for process-window analysis, we perform a 1000-iteration Monte Carlo (MC) simulation for each in-line sample. Consequently, for a given Parylene 3D KiSS array i , the measured geometric feature sizes (l , w , and s) are replicated into 3×1000 matrices, while the unknown λ parameter is stochastically

sampled 1000 times within its allowable tolerance window. The resulting hybrid data set is fed into the reliable ML models to predict k_{eff} and $\varepsilon_s^{\max}$.

Adopting the same mechanical performance specification limits defined earlier, we compute the qualification probability P_{qual} for each sample immediately after photolithography, which is well before λ is fixed in later steps. A sample with high P_{qual} indicates robust l , w , and s combinations with low λ -related process risks, whereas a sample with low P_{qual} is defined as a high-risk sample. For instance, sample i in Fig. 5 yields $P_{\text{qual}} = 15.3\%$, far below the target threshold, signaling the need for extra adjustments of the fabrication deviations of λ in subsequent process.



Repeating this quasi-Monte Carlo (QMC) procedure across all sites on the wafer, wafer-level assessment of qualification probability for Parylene 3D KiSS array can be achieved. The proposed real-time assessment methodology of fabrication deviation induced mechanical performance variations enables in-line wafer qualification probability decision-making, empowering high-consistency manufacturing of Parylene 3D KiSS arrays.

Conclusions

This study demonstrates a machine learning-driven framework that effectively addresses mechanical performance variations arising from fabrication-induced geometric deviations in Parylene 3D Kirigami-like stretchable silicon

array (Parylene 3D KiSS array). By integrating dimensionality reduction and data-driven machine learning modeling, the framework predicts two key mechanical performance metrics of Parylene 3D KiSS arrays with high fidelity: an accuracy of 97.27% for the equivalent Young's modulus and 95.51% for the maximum elastic stretchability. Furthermore, the SHapley Additive exPlanations analysis (SHAP) further quantifies the relative importance of the feature sizes in the Parylene 3D Kirigami-like microstructures, providing interpretable insights into the design-performance relationship. The proposed methodology enables real-time prediction of amplified mechanical performance variations caused by the feature size deviations of the miniaturized microstructure during MEMS microfabrication processes.

This establishes a foundation for high-consistency manufacturing in silicon-based stretchable electronics through design-for-manufacturability approaches. These advances facilitated the transition from lab-scale prototypes to standardized silicon-based stretchable electronics production while maintaining wafer-level benefits.

Methods

Fabrication process of the Parylene 3D KiSS arrays

The fabrication of Parylene 3D KiSS arrays begins on 4-inch silicon wafers. The Parylene 3D Kirigami-like microstructures are firstly patterned via photolithography process where the feature sizes are defined and obtained by optical microscope. Then, DRIE is performed to etch the silicon wafer to prepare the Si mold. The Parylene is consequently deposited conformally to fill into the silicon mold, and excess Parylene film on top the wafer is removed from the silicon wafer by oxygen reactive ion etching until the Si top surface is exposed. A second photolithography process is conducted to protect the silicon island, followed by silicon DRIE to remove the residue silicon inside the Parylene 3D Kirigami-like microstructures. Finally, the entire wafer is grinded from the backside until the Parylene 3D Kirigami-like microstructures is fully exposed from the backside.

Tensile tests

The prepared Parylene 3D KiSS arrays are subjected to uniaxial tensile tests to extract the equivalent Young's modulus and the maximum elastic stretchability. The whole tensile process is done on a homemade tensile test platform. The tensile speed is set to 3 mm/min, and the tensile force data is recorded every 0.01 s. The samples are attached to the gadgets by tapes for the tensile testing. The initial width W_0 is measured before stretching. The force-displacement curves are directly obtained from the uniaxial tensile tests. The W_0 and the cross-sectional area (A , calculated by the thickness t times length L) are used to calculate the strain-stress curves. The equivalent Young's modulus and the maximum elastic stretchability are then extracted from the strain-stress curves.

ML modeling

In this study, we construct four multilayer-perceptron (MLP) architectures and four CatBoost architectures for mechanical performance modeling. Each model undergoes 5-fold cross-validation to maximize robustness and predictive accuracy. All input features are rescaled to the [0,1] interval using min-max normalization, which improves numerical stability and speeds up convergence. The weight-decay regularization with $h = 0.001$ is imposed across all eight models to enhance generalization capability and mitigate overfitting. The Sigmoid activation function^{36,37}, widely adopted for regression tasks, is

implemented in all four MLP models. The four MLP models are trained via the Adam optimizer, with a variable learning rate strategy starting at 0.001. The dataset is stratified into 77% training and 23% testing subsets, and an early-stopping criterion selects the optimal epoch to prevent over-training.

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Conflict of interest

W.W. is an editor for this journal, no other author has reported any competing interest.

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